Dengue Outbreak Early Warning for Sri Lanka: Negative Binomial and RuleFit with Explainable AI

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Abstract

Dengue outbreaks in Sri Lanka are usually recognised only once hospitals are already filling up. In this project I built an explainable early-warning system for the country's 26 districts and health areas. I combined 25,090 weeks of official case counts (2007–2025) with weather data and applied two algorithms not covered in our lectures. Negative Binomial regression forecasts weekly cases together with a range of uncertainty. On the 2022–25 test years it achieved $R^2 = 0.80$, and 93.5% of actual counts fell within its 90% ranges. RuleFit flags likely outbreak weeks using a short list of readable rules, reaching a PR-AUC of 0.43 against a base rate of 0.088. Neither beat Random Forest or XGBoost outright, but both came close, and both are far easier to explain. Five explanation methods (incidence rate ratios, rules, SHAP, partial dependence and LIME) agree that recent cases drive most predictions. They also show that humidity and warmth two to four weeks earlier, and rainfall six to nine weeks earlier, increase risk, which fits the mosquito's life cycle. A Streamlit dashboard lets users try what-if scenarios and see the reasons behind every forecast.

1. Problem definition and dataset

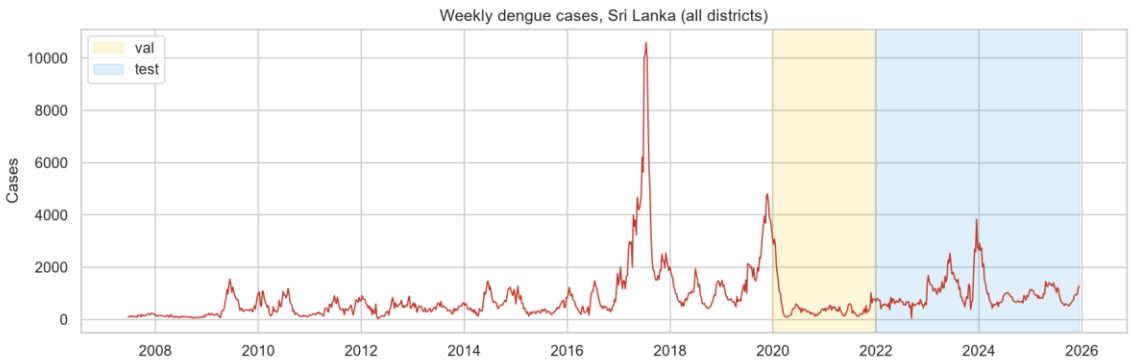
1.1 The problem and why it matters

Almost every year, dengue fills hospital wards somewhere in Sri Lanka. In 2017 the weekly reports recorded about 175,000 cases, and in the worst week of that epidemic about 10,500 people were notified in seven days (Figure 1). Health officers usually find out an outbreak has started only after the cases are already arriving. By then, fogging, clean-up drives and extra beds are a step behind.

I wanted to see whether routinely published data could give officers a short warning, and a reason for it. The project therefore has two connected tasks:

- **Task A (regression):** how many dengue cases will district *d* report in week *t*?
- **Task B (classification):** will week *t* be an outbreak week for district *d*?

The "reason" part is as important as the forecast. A health officer is far more likely to act on a warning such as "cases have been rising for two weeks and it has been very humid" than on an unexplained number.



1.2 Where the data came from

I did not find a ready-made dataset that combined dengue cases and weather at district level, so I built one from two public sources.

The case counts come from the **Weekly Epidemiological Reports** that the Epidemiology Unit of the Ministry of Health publishes as PDFs. Parsing nineteen years of PDFs by hand would have been error-prone. Fortunately, the R package denguedatahub (v4.1.0) already contains these reports in a tidy table. My script `collect_data.py` downloads that package from CRAN and extracts the Sri Lankan weekly table: 25,766 rows covering 26 districts and health areas from December 2006 to December 2025. Kalmunai is reported separately from Ampara as its own health area, so I kept it as a 26th unit.

The weather data come from the free **Open-Meteo historical weather API**, which serves ERA5 reanalysis data. For each district I chose an approximate centre point and downloaded daily rainfall, mean temperature and mean relative humidity. This gave 6,997 days for each of the 26 locations, 181,922 rows in total.

Source	What it contains	How I collected it
Epidemiology Unit weekly reports (via denguedatahub)	Weekly notified dengue cases, 26 districts/areas, 2006–2025	Downloaded automatically from CRAN by `collect_data.py`
Open-Meteo / ERA5	Daily rain (mm), temperature (°C), humidity (%)	One API request per district centre point

1.3 Features and target

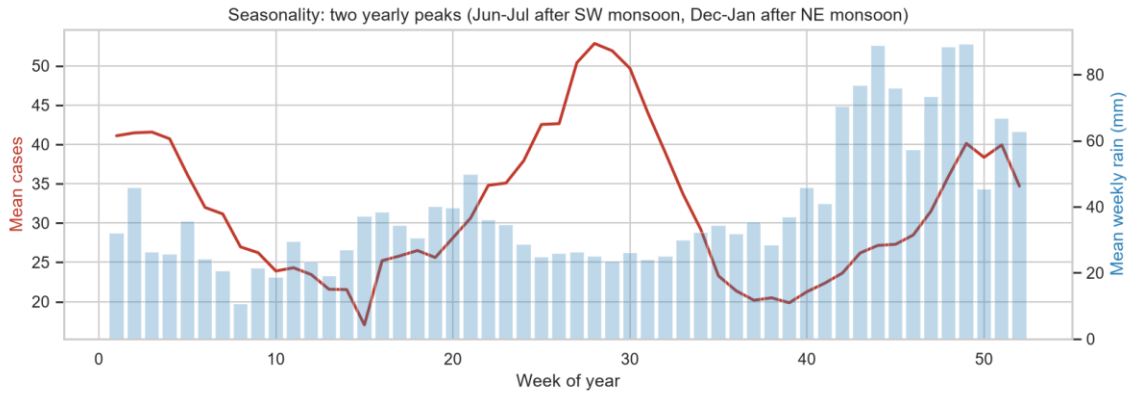
Each row in the final dataset is one district in one week. A rule I kept to throughout was that every feature must be something an officer would already know *before* that week starts. Otherwise the model would be cheating.

Group	Features	Why I included them
Recent cases	log(1 + cases) 1 and 2 weeks ago; log of the 4-week average	Dengue spreads from person to mosquito to person, so recent cases predict new ones

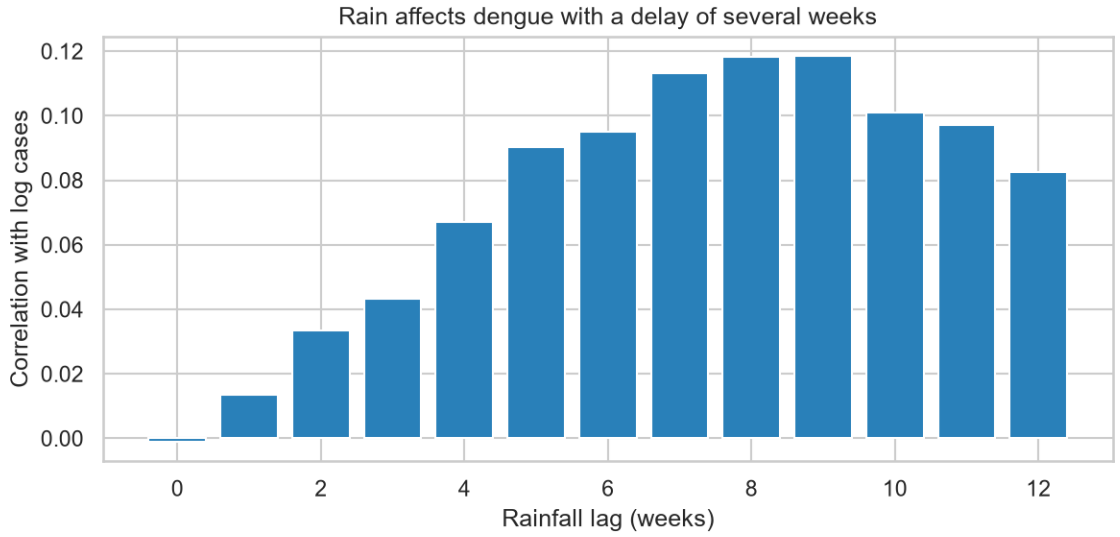
Group	Features	Why I included them
Weekly weather	Rain, temperature and humidity 1–4 weeks ago	These drive mosquito breeding and survival
Long rain lags	Rain 8 weeks ago; 4-week rain totals ending 1 and 6 weeks ago	Added after the EDA showed rain matters most 7–9 weeks later (Figure 3)
Outbreak context	The district's outbreak threshold; last week's cases relative to it	How close a district already is to an outbreak
Season and place	Week of year (as sine/cosine), monsoon period, district	Seasonal peaks and baseline differences between districts

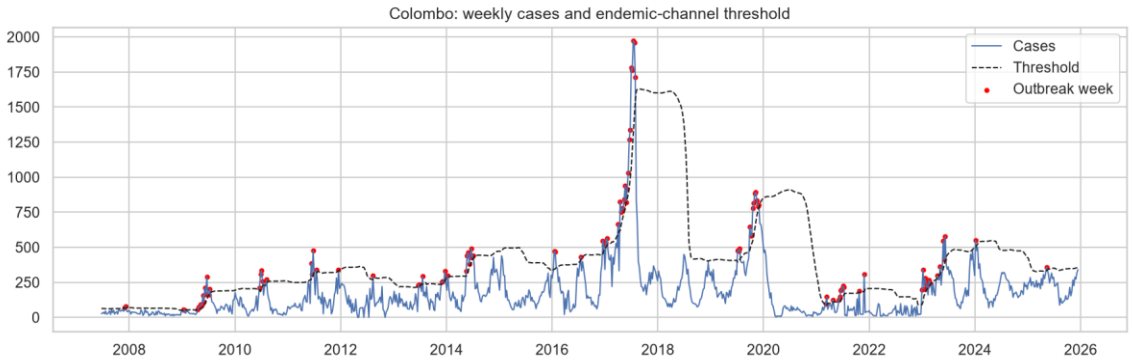
For **Task A** the target is simply the weekly case count, which ranges from 0 to 2,631.

For **Task B** I needed a definition of an "outbreak". I used the endemic-channel idea that is common in disease surveillance. A week is an outbreak week if its cases are above the district's mean plus two standard deviations over the previous 52 weeks. Because the threshold uses only past weeks, the label never peeks into the future. Figure 4 shows how this works for Colombo.



The exploratory analysis changed my feature set. I first assumed that rain from the last month would matter most. When I measured the correlation between rainfall k weeks ago and this week's cases (after removing differences between districts), it kept rising until about **7–9 weeks** (Figure 3). That makes sense biologically: eggs need time to hatch, adult mosquitoes need time to become infectious, and people need time to fall ill. So I added the longer rain lags.



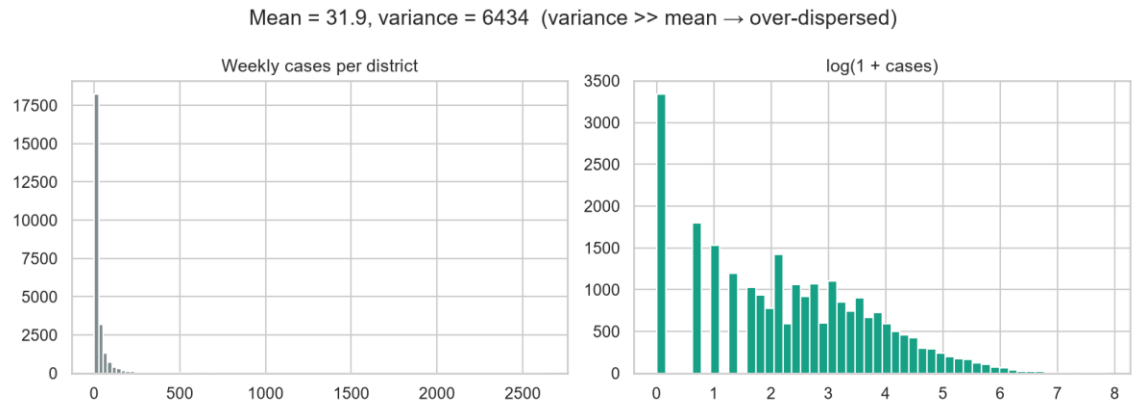


1.4 Size and preprocessing

After cleaning, the modelling dataset has **25,090 district-weeks** from June 2007 to December 2025, with **49 features**. Of these weeks, **2,494 (9.9%)** are outbreak weeks.

The main preprocessing steps were:

1. **Cleaning.** The source file mixed two date formats, so I parsed them explicitly. I then removed rows without a case count and dropped duplicate district-weeks.
2. **Turning daily weather into weekly values.** I matched each day to the report week whose start and end dates contain it, then summed rainfall and averaged temperature and humidity.
3. **Building lag features.** All lags were calculated within each district, so one district's history never leaks into another's. I log-transformed the case features because the counts are extremely skewed: the mean is 31.9 but the variance is 6,434 (Figure 5).
4. **Encoding and scaling.** District and monsoon were one-hot encoded. Numeric features were standardised using statistics from the training data only. For RuleFit I kept the raw units so that its rules would read as "humidity > 78%" rather than "humidity > 0.43 SD".
5. **Handling missing values.** The outbreak threshold needs at least half a year of history, so the first 26 weeks of each district had nothing to be compared against. I dropped these 676 warm-up rows (26 weeks × 26 districts) rather than fill them with guesses.



1.5 Ethical use of the data

Both datasets are public and aggregated. The dengue data are weekly counts per district. They contain no names, ages, addresses or any other information about individual patients, so no consent was required and no one can be re-identified. The weather data are gridded estimates, not personal data at all. Both sources allow academic reuse (denguedatahub is GPL-3 and Open-Meteo is CC BY 4.0), and I cite them in the references.

2. Selection of new machine learning algorithms

I used one new algorithm for each task: **Negative Binomial regression** for the case counts and **RuleFit** for the outbreak alerts. Neither was covered in our lectures, and neither is a deep learning model. I didn't pick them just for being new. Each one fixes a specific problem that the standard models struggle with on this kind of data.

2.1 Negative Binomial regression (Task A)

Negative Binomial regression is a generalised linear model for count data. It assumes the weekly count Y follows a Negative Binomial distribution with mean μ, and it links that mean to the features through a logarithm:

$$\log \mu = \beta_0 + \sum_j \beta_j x_j, \qquad \mathrm{Var}(Y) = \mu + \alpha \mu^2$$

The important part is the variance formula. The extra term $\alpha\mu^2$ lets the spread of the counts grow faster than their average. That is exactly what happens with dengue: a quiet district might report 2 to 5 cases a week, while a district in an epidemic can jump from 200 to 600. For this dataset, the maximum-likelihood estimate of α is **0.308**.

I chose this model for three reasons.

First, it suits the shape of the target. The weekly counts have a variance about 200 times their mean (Figure 5). Ordinary linear regression assumes a constant variance and can even predict negative cases. A Poisson model forces the variance to equal the mean, which is far from true here. Negative Binomial is the standard choice in epidemiology for this kind of over-dispersed count.

Second, it predicts a whole distribution, not just one number. From each forecast I can read off a 90% range of plausible case counts. For planning hospital beds, "probably 150 to 1,100 cases" is more useful

than "535 cases". On the test data, 93.5% of the actual counts fell inside these 90% ranges, so the uncertainty is honest.

Third, its coefficients have a clear meaning. exp(β) is an *incidence rate ratio* (IRR): the factor by which expected cases multiply when a feature increases. An IRR of 1.08 for humidity, for example, means 8% more expected cases.

2.2 RuleFit (Task B)

RuleFit was proposed by Friedman and Popescu (2008). It works in two stages:

1. It grows many shallow decision trees and turns every path from the root to a node into a yes/no rule, for example *"humidity two weeks ago > 78% AND last week's cases > 68% of the threshold"*.
2. It fits a logistic model with an L1 (Lasso) penalty on all of these rules, together with the original features. The penalty sets most rule weights to zero, so only a small set survives.

The final model is a weighted sum of readable rules and a few linear terms:

```
F(x) = \beta_0 + \sum_{k} a_k \, r_k(x) + \sum_j b_j \, x_j
```

Here $r_k(x)$ is 1 if rule *k* is true for that week and 0 otherwise. My fitted model kept **31 rules and 7 linear terms**.

I chose RuleFit because an outbreak alert has to be trusted before anyone acts on it. A Medical Officer of Health can read a rule such as "cases two weeks ago were above 5 and last week was already at two-thirds of the threshold" and decide whether it makes sense locally. That is not possible with a forest of hundreds of trees. RuleFit can still capture interactions, such as humidity mattering only when cases are already climbing, which a plain logistic regression cannot. Because the model *is* the list of rules, the explanation is exact rather than an approximation added afterwards.

2.3 How they differ from the models in the lectures

Lecture model	Compared with	Main difference
Linear Regression, Ridge, Lasso	Negative Binomial	Linear regression assumes normally distributed errors with constant variance. Negative Binomial models non-negative whole numbers, lets the variance grow with the mean ($\mu + \alpha\mu^2$), and gives prediction intervals.
Logistic Regression	Negative Binomial	Both are GLMs, but logistic regression predicts a yes/no outcome. Negative Binomial predicts a count and estimates an extra dispersion parameter.
Decision Trees, Random Forest, Gradient Boosting, XGBoost	RuleFit	RuleFit uses trees only to *suggest* rules, then keeps a short weighted list with Lasso. The result is a few dozen readable rules, not hundreds of trees voting.

Lecture model	Compared with	Main difference
Lasso, Logistic Regression	RuleFit	RuleFit applies the Lasso penalty to automatically discovered interaction rules as well as the raw features, so it can learn thresholds and combinations that a linear model cannot.
Explainable Boosting Machine (briefly mentioned in the XAI lecture)	Both	EBM learns a smooth curve for each feature separately. RuleFit learns explicit combinations of conditions, and Negative Binomial gives a full probability distribution for the count.

To judge whether the new models were actually worth using, I compared each against three lecture models trained on the same features and the same data split (Section 3).

3. Model training and evaluation

In short: on four years of data the models had never seen (2022–2025), Negative Binomial explained about 80% of the variation in weekly cases, and its 90% ranges contained the true count 93.5% of the time. RuleFit reached a PR-AUC of 0.43, almost five times better than guessing. Both new models finished close to the strongest lecture models.

3.1 How I split the data

Because the aim is to forecast the *future*, I split the data by time instead of shuffling it. With a random split, a test week would usually have its neighbouring weeks in the training set, and the scores would look much better than they would be in real use.

Split	Years	Rows	Share of outbreak weeks	Used for
Train	2007–2019	16,978	10.7%	Fitting the models
Validation	2020–2021	2,730	7.6%	Choosing hyperparameters and alert cut-offs
Test	2022–2025	5,382	8.8%	Final evaluation, used only once

I followed the same procedure for every model. I tried each candidate setting on the training years and scored it on the validation years. Then I refitted the best setting on 2007–2021 and evaluated it once on 2022–2025. The feature scaler was always fitted on the training portion only, so no information from the test years leaked into training.

3.2 Hyperparameter choices

Model	What I searched	What was selected	How it was selected
Negative Binomial	Dispersion α from 0.005 to 0.5, and maximum likelihood	α = 0.308	Maximum likelihood (see below)

Model	What I searched	What was selected	How it was selected
Random Forest (counts)	max_depth {8, 12, none} × min_samples_leaf {5, 20}; 300 trees	depth 12, leaf 20	Validation Poisson deviance
XGBoost (counts)	max_depth {3, 5} × learning rate {0.03, 0.1}; 400 trees, Poisson objective	depth 5, rate 0.03	Validation Poisson deviance
RuleFit	max_rules {10, 20, 40} × tree_size {3, 4}	40 rules, tree size 3	Validation PR-AUC (0.571)
Logistic Regression	C {0.01, 0.1, 1}; balanced class weights	C = 0.01	Validation PR-AUC (0.537)
Random Forest (outbreaks)	max_depth {8, 12} × min_samples_leaf {5, 20}; balanced	depth 12, leaf 20	Validation PR-AUC (0.539)
XGBoost (outbreaks)	max_depth {3, 5} × learning rate {0.03, 0.1}; positive-class weight 8	depth 3, rate 0.1	Validation PR-AUC (0.561)

One choice needs explaining. I first tuned α by grid search on validation Poisson deviance, and it kept picking the smallest value I offered (0.005), which is practically a Poisson model. I realised this is a bias in the metric itself. Poisson deviance is the Poisson model's own loss function, so it naturally favours Poisson-like fits. I therefore estimated α by maximum likelihood, which is the standard method and gave 0.308. The grid results are still saved in `tuning_count.csv`.

For each classifier I also chose the probability cut-off for raising an alert on the validation years, picking the value that gave the best F1 score.

3.3 Performance metrics

For the counts, I report MAE and RMSE (in cases per week) and R^2. I added the **mean Poisson deviance**, a fairer error measure for counts than squared error. For Negative Binomial, I also checked **90% interval coverage**: the share of weeks where the true count fell inside the predicted range.

For outbreaks, accuracy would be misleading. Only 8.8% of test weeks are outbreaks, so a model that always says "no outbreak" scores 91% accuracy while being useless. My main metric was therefore **PR-AUC**, which focuses on how well the model finds the rare positive weeks. I also report ROC-AUC, F1, precision and recall at the chosen cut-off.

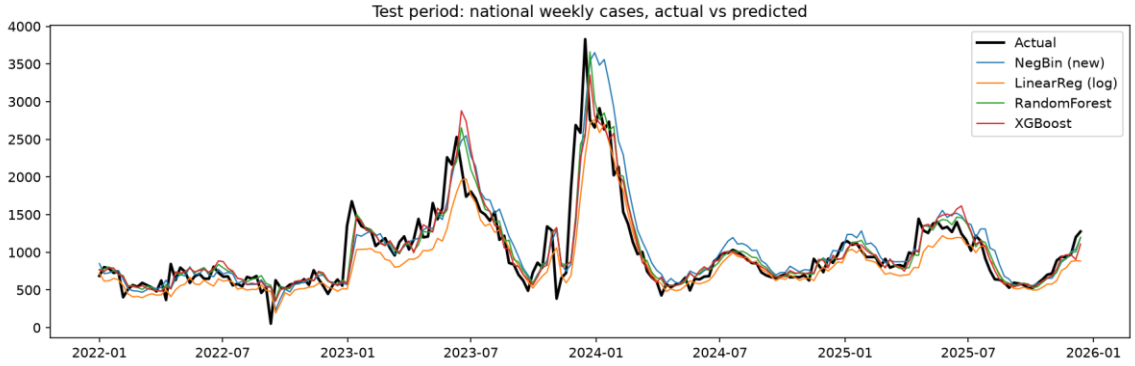
3.4 Results

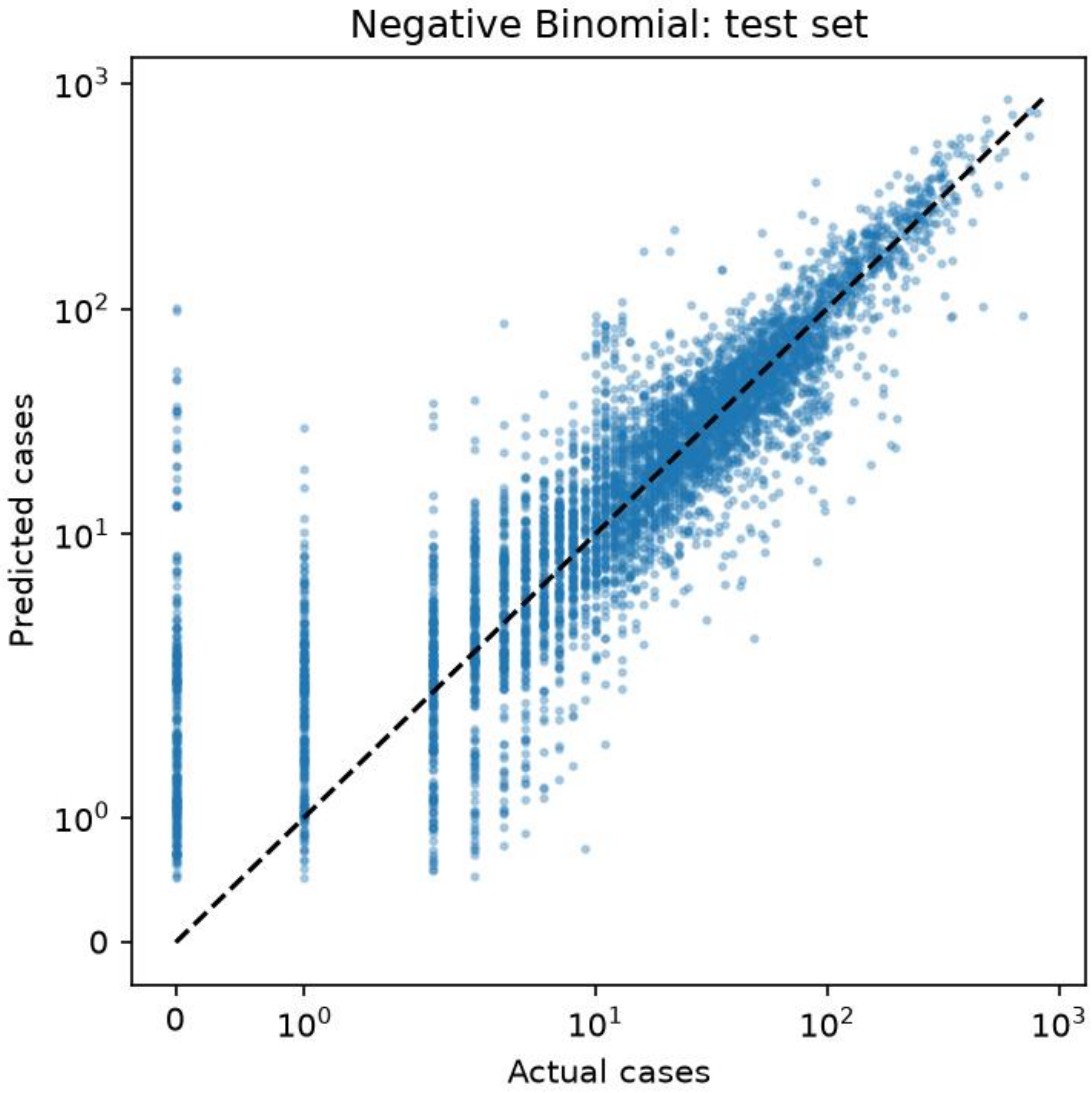
Task A: weekly case counts on the test set (2022–2025)

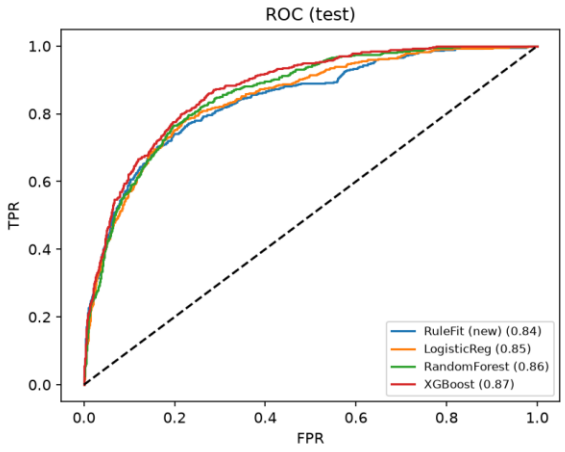
Model	MAE	RMSE	R^2	Poisson deviance	90% interval coverage
Negative Binomial (new)	12.81	27.94	0.795	8.68	**0.935**
Linear Regression (log target)	11.94	26.71	0.813	9.15	–
Random Forest	**11.75**	26.12	0.821	**7.99**	–
XGBoost (Poisson)	11.84	**26.08**	**0.822**	8.03	–

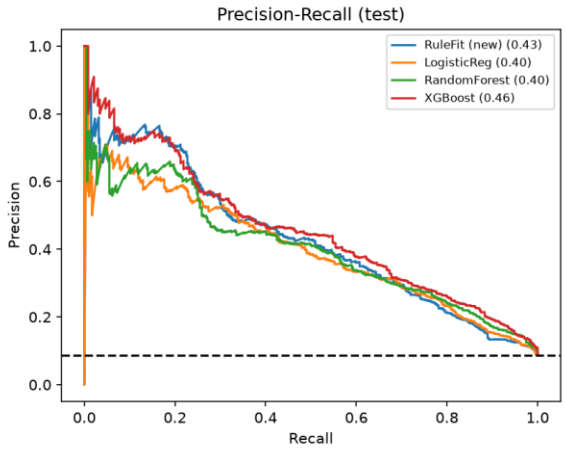
Task B: outbreak weeks on the test set (471 outbreak weeks out of 5,382)

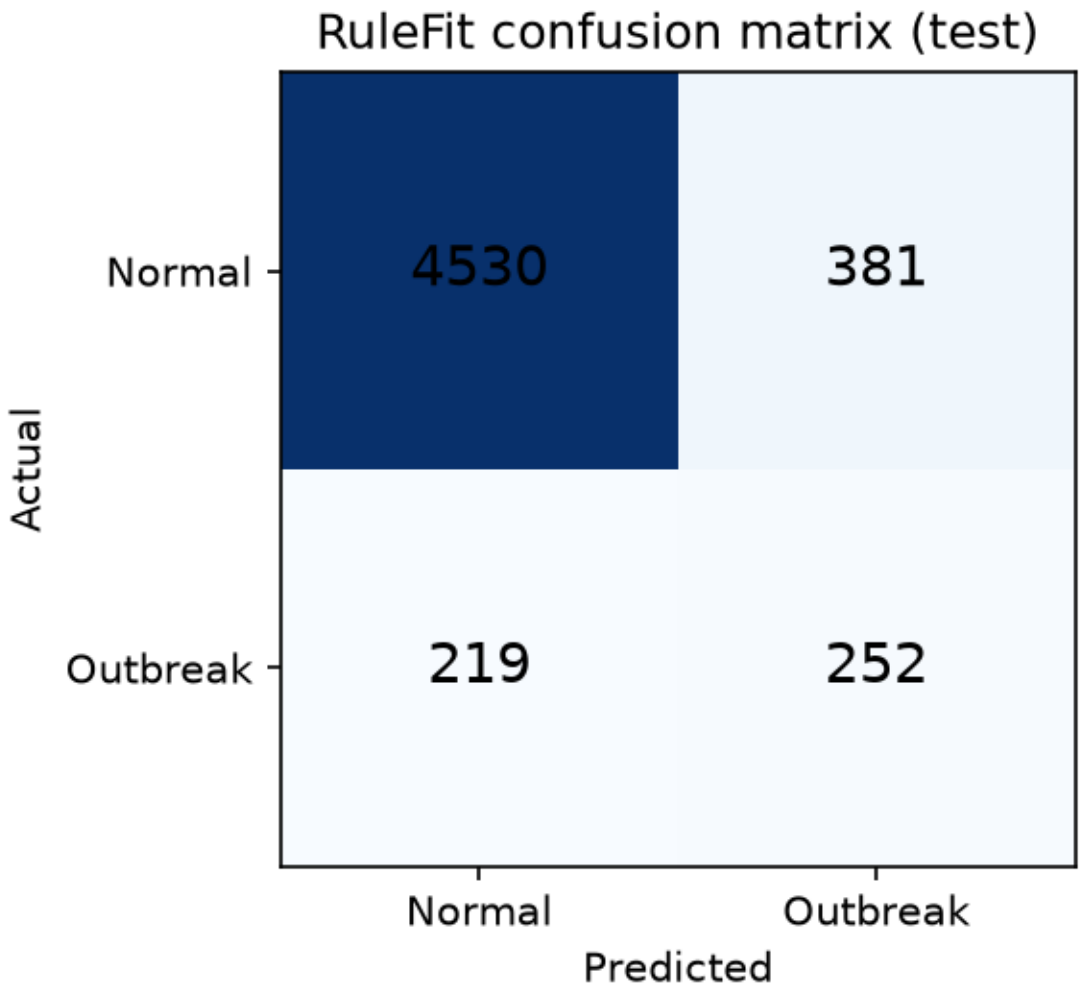
Model	ROC-AUC	PR-AUC	F1	Precision	Recall
RuleFit (new)	0.841	0.429	0.457	**0.398**	0.535
Logistic Regression	0.846	0.397	0.430	0.343	0.577
Random Forest	0.859	0.404	0.442	0.362	0.567
XGBoost	**0.873**	**0.456**	**0.462**	0.364	**0.631**




Negative Binomial: test set
Predicted cases
Actual cases
0
10^0
10^1
10^2
10^3


ROC (test)
TPR
FPR
RuleFit (new) (0.84)
LogisticReg (0.85)
RandomForest (0.86)
XGBoost (0.87)


Precision-Recall (test)
Precision
Recall
RuleFit (new) (0.43)
LogisticReg (0.40)
RandomForest (0.40)
XGBoost (0.46)



3.5 What the results tell us

All four count models follow the epidemic curve well, including the large waves in mid-2023 and early 2024 (Figure 6), with R^2 values between 0.80 and 0.82. Negative Binomial is about one case per week less accurate on average than the tree ensembles. I think that is a reasonable price, because it is the only model that says how uncertain each forecast is, and its uncertainty proved reliable (93.5% coverage for a 90% interval). It is also slightly cautious: its ranges are a little wider than needed.

Figure 6 also reveals a weakness that every model shares: they react to sharp turns about one week late. Since recent cases are the strongest predictor, the models recognise a surge only once it has begun. I come back to this in the discussion.

For outbreak detection, RuleFit had the **highest precision of the four classifiers (0.40)**: when it raised an alert, it was right more often than the others. It caught 252 of the 471 outbreak weeks (53.5%), with 381 false alarms (Figure 9). XGBoost was about 0.03 better on PR-AUC and ROC-AUC. However, XGBoost's

400 trees cannot be inspected, while RuleFit's 38 terms can be read in a few minutes. For a public-health alert I consider that trade-off worth making.

The biggest single improvement came from feature engineering rather than model choice. Adding a feature for how close last week's cases were to the outbreak threshold lifted every classifier's validation PR-AUC from about 0.24 to above 0.53. In other words, the most useful information is not the weather on its own, but how near a district already is to tipping over.

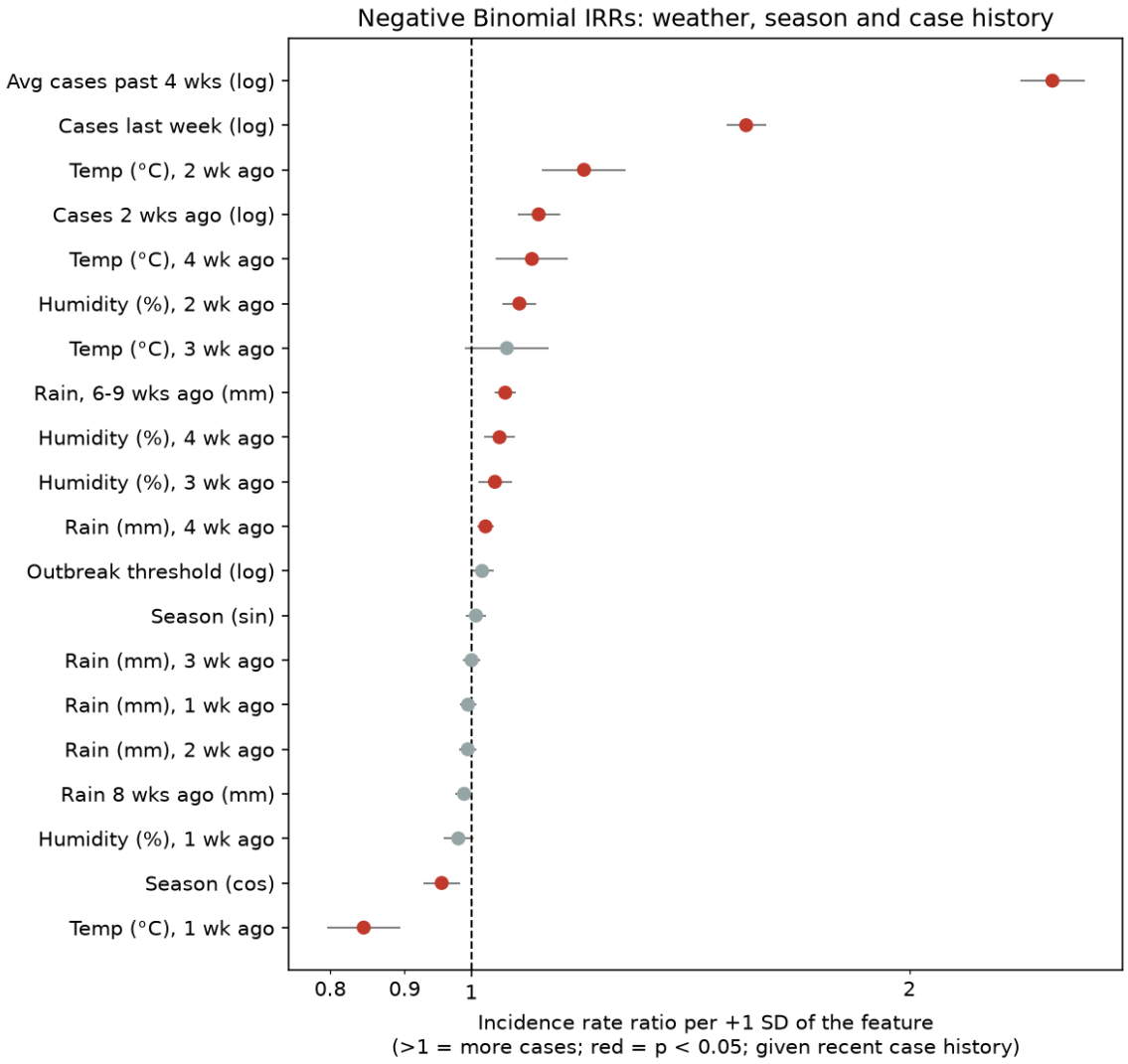
4. Explainability and interpretation

I ended up using five explanation methods. One would have met the guidelines, but I wanted to see whether different methods would agree, because if they didn't, I couldn't really trust any of them. They told a consistent story. **Recent case momentum drives most predictions. Weather adds a smaller but sensible signal: humid, warm weeks 2–4 weeks earlier, and rainy weeks about two months earlier, push expected cases up.**

Method	What it explains	Applied to	Figures
Incidence rate ratios	The whole model (built in)	Negative Binomial	10–11
Learned rules	The whole model (built in)	RuleFit	12
SHAP	Every prediction, and the overall picture	Negative Binomial (exact), RuleFit (permutation), XGBoost (TreeSHAP)	13–14
Partial dependence plots	The average effect of one feature	Negative Binomial and XGBoost	15
LIME	One single prediction	RuleFit	16

4.1 What Negative Binomial learned: incidence rate ratios

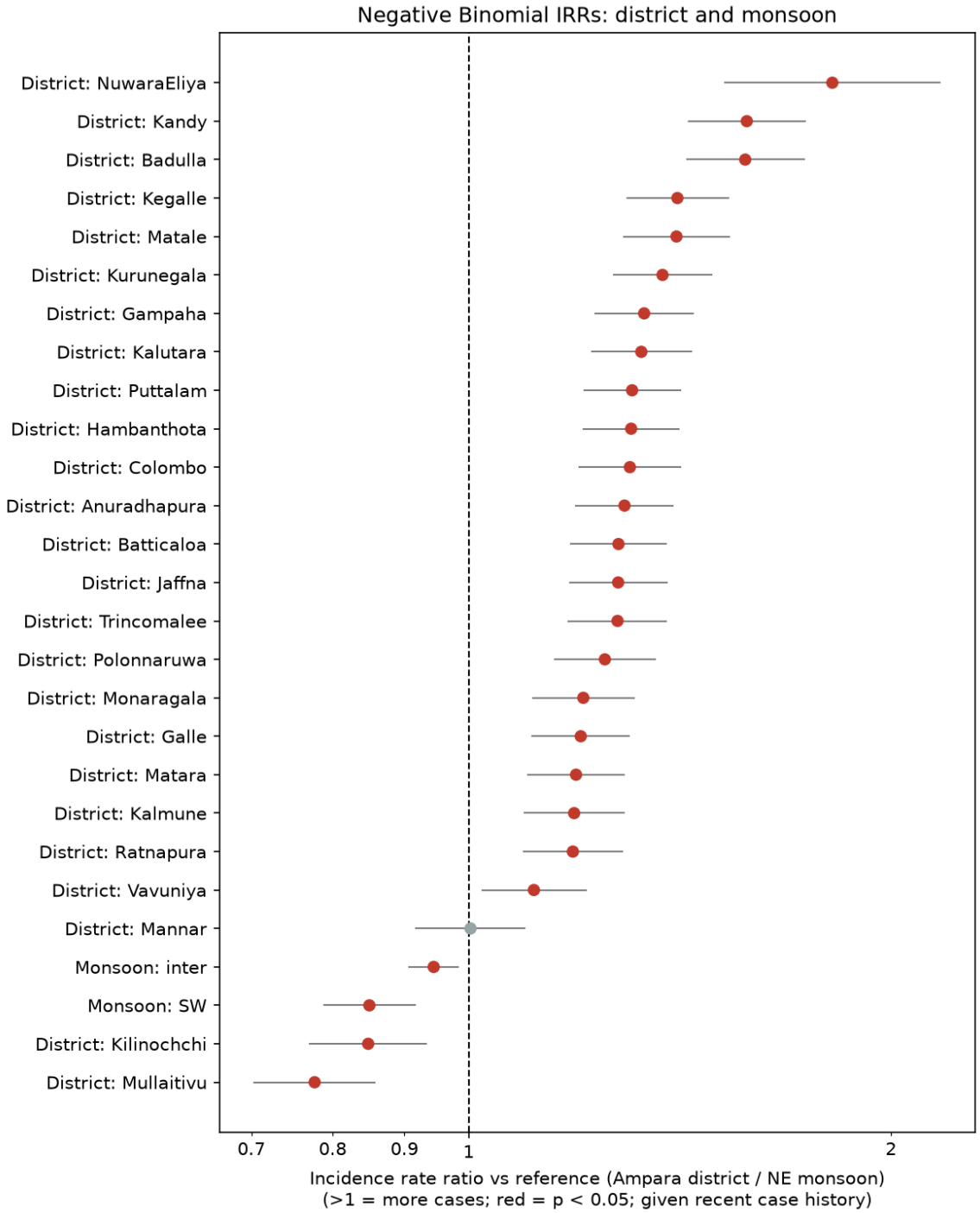
Each point in Figure 10 shows how much expected cases change when one feature rises by one standard deviation, with everything else held fixed. A value of 1 means no effect; red points are statistically significant ($p < 0.05$).



Feature (+1 SD)	IRR	What it means
Average cases over the past 4 weeks (log)	2.51	By far the strongest effect: sustained transmission leads to more cases
Cases last week (log)	1.55	Short-term momentum
Mean temperature 2 weeks ago (+2.6 °C)	1.20	Warmth speeds up how quickly the virus develops inside the mosquito
Cases 2 weeks ago (log)	1.11	More momentum
Humidity 2 weeks ago (+7.8%)	1.08	Mosquitoes live longer in humid air
Rain over 6–9 weeks ago (+136 mm)	1.055	New breeding sites created about two months before

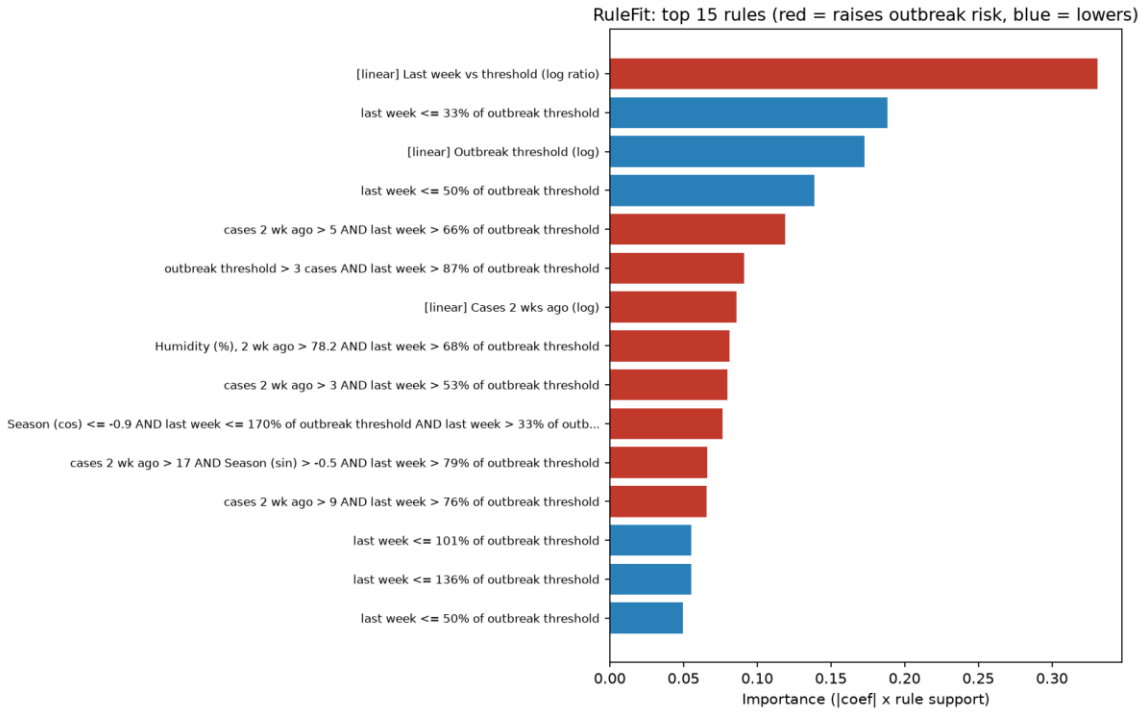
Feature (+1 SD)	IRR	What it means
Temperature 1 week ago (+2.6 °C)	0.84	Negative, which surprised me; explained in Section 4.6
Rain 1–3 weeks ago	about 1.00 ($p > 0.4$)	No clear short-term effect

The district effects in Figure 11 are relative to Ampara, and they apply *after* recent cases are taken into account. For example, Nuwara Eliya's IRR of 1.82 does not mean it has the most dengue. It means that, for the same number of recent cases, its numbers tend to grow faster. The south-west monsoon months have about 15% fewer expected cases than the north-east monsoon months, again at equal case history.



4.2 What RuleFit learned: its rules

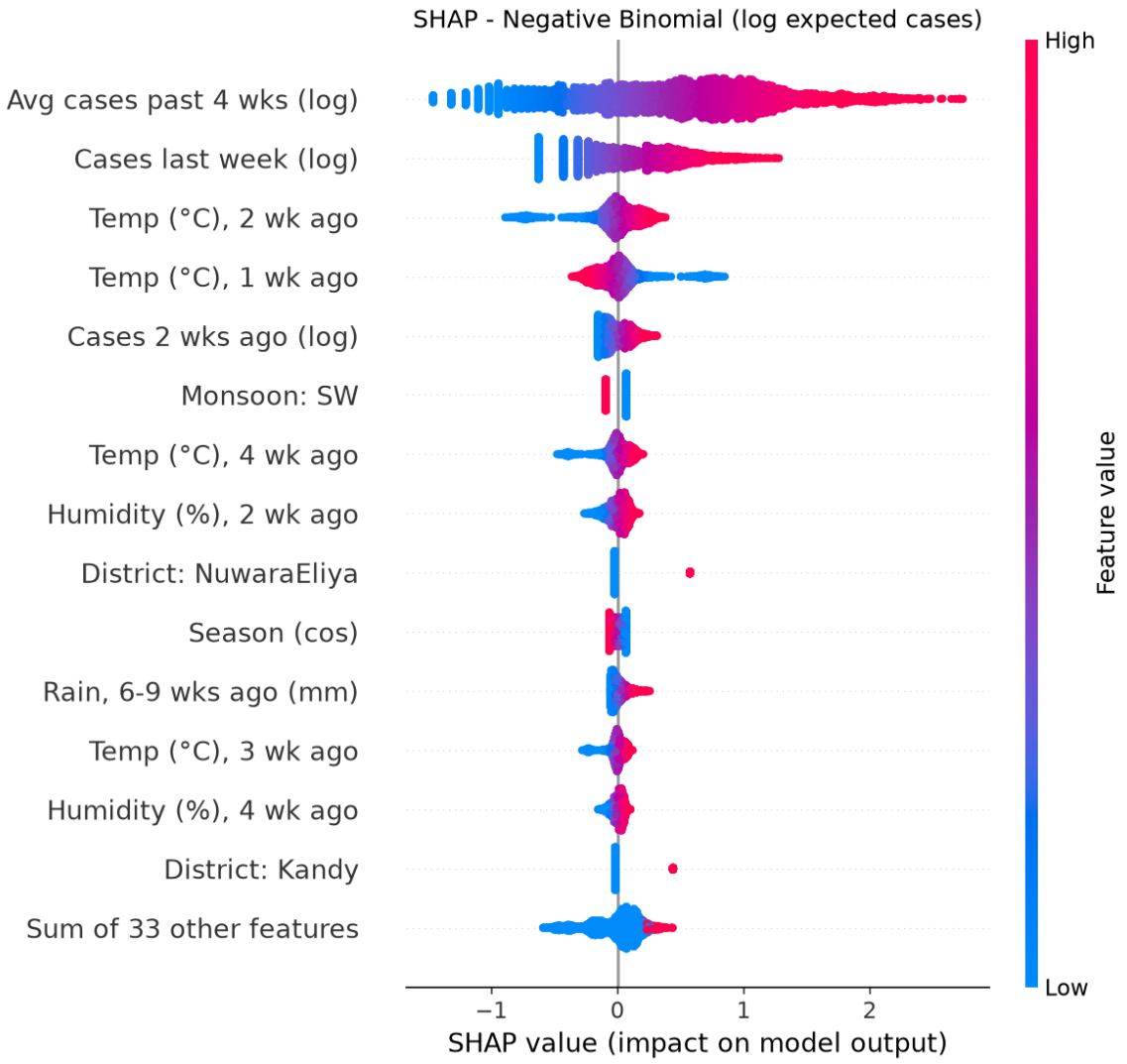
With RuleFit the model's rules are the explanation, so nothing needs to be approximated. Figure 12 shows the 15 most important terms, which I translated from log units back into plain language ("last week ≤ 33% of the threshold").

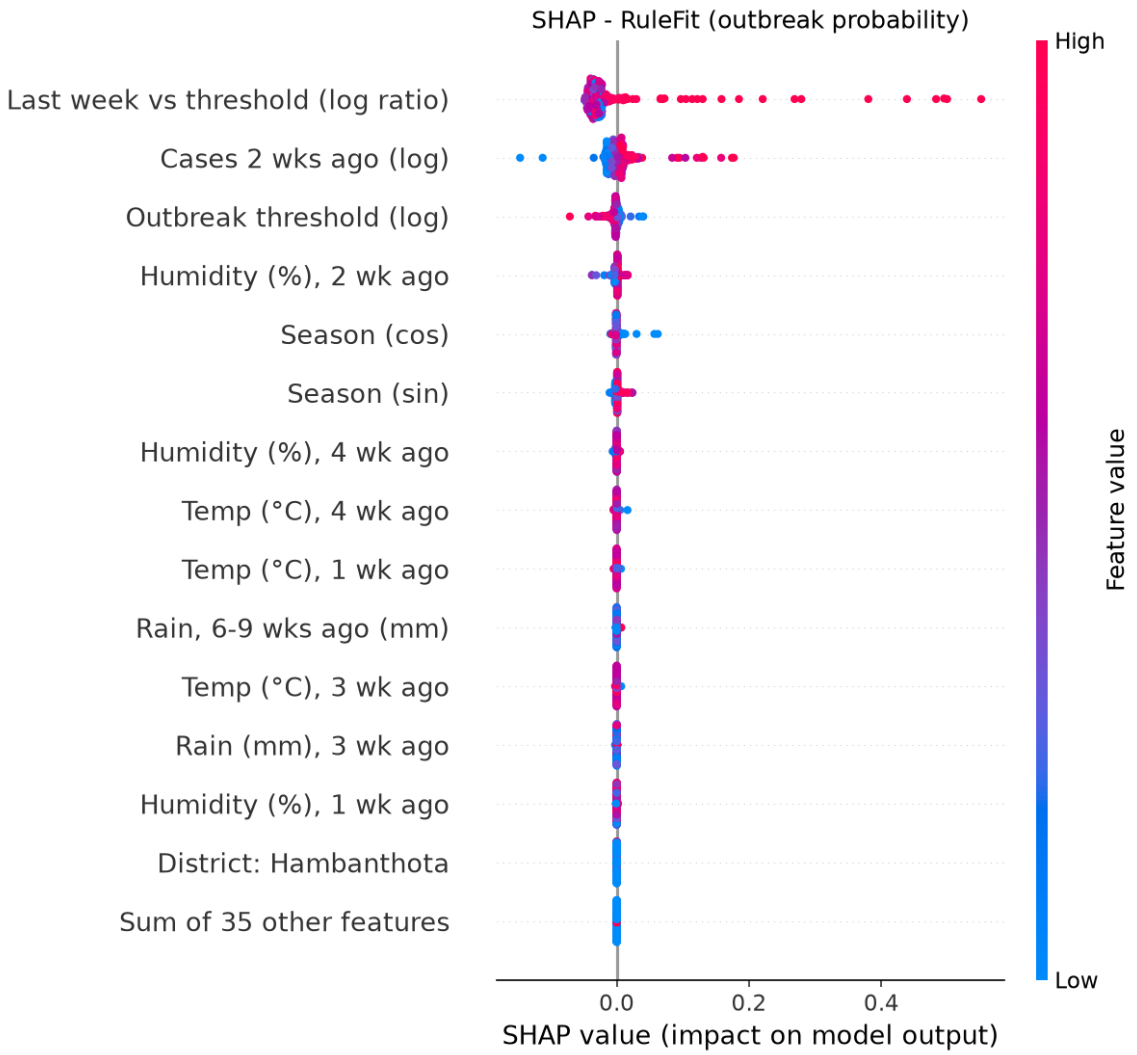


The rules that raise risk almost all combine two things: cases have been building for a couple of weeks, and last week was already close to the threshold. A typical example is *"cases two weeks ago > 5 AND last week > 66% of the threshold"* (weight +0.32). Weather appears mainly inside combinations, such as *"humidity two weeks ago > 78% AND last week > 68% of the threshold"* (+0.23). I found this the most interesting result, because it says humidity matters only once transmission is already under way, which a linear model could never express. The rules that lower risk are the reassuring ones: if a district was below a third or half of its threshold last week, an outbreak next week is very unlikely.

4.3 SHAP

SHAP splits a single prediction into contributions from each feature, which add up to the final output. For Negative Binomial these values can be calculated exactly on the log scale, $\phi_j = \beta_j(x_j - E[x_j])$. For RuleFit I used the permutation explainer on 300 test weeks, because it is model-agnostic.

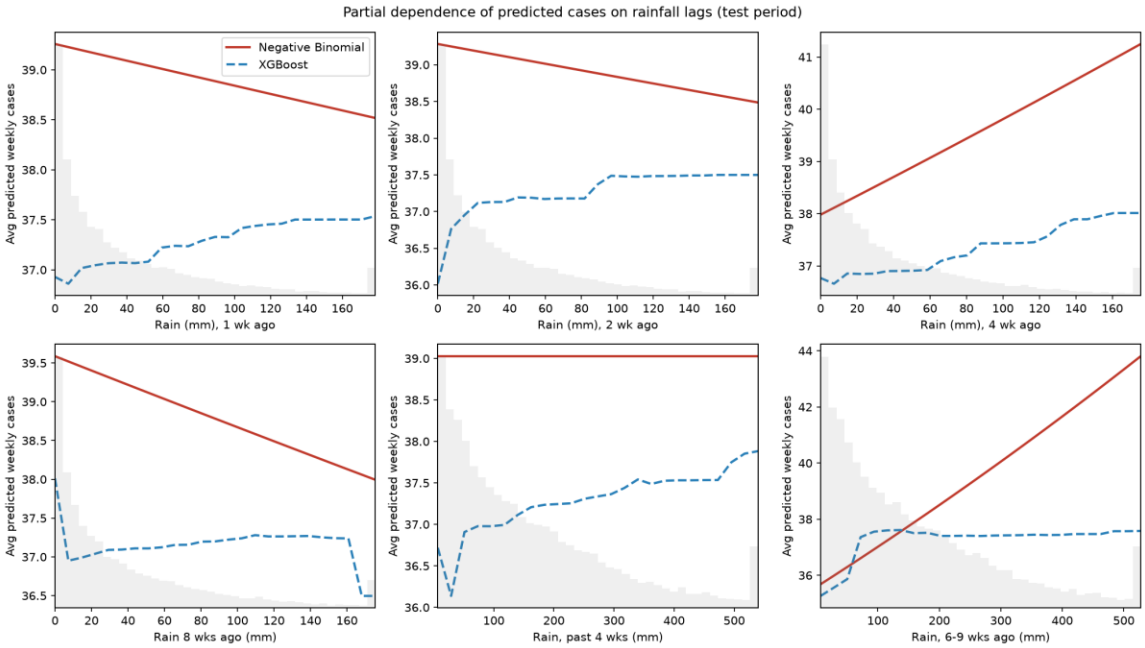

SHAP - Negative Binomial (log expected cases)
Avg cases past 4 wks (log)
Cases last week (log)
Temp (°C), 2 wk ago
Temp (°C), 1 wk ago
Cases 2 wks ago (log)
Monsoon: SW
Temp (°C), 4 wk ago
Humidity (%), 2 wk ago
District: NuwaraEliya
Season (cos)
Rain, 6-9 wks ago (mm)
Temp (°C), 3 wk ago
Humidity (%), 4 wk ago
District: Kandy
Sum of 33 other features
−1
0
1
2
SHAP value (impact on model output)
High
Feature value
Low



The three models I explained with SHAP (Negative Binomial, RuleFit and the XGBoost baseline) all put recent case history and the outbreak threshold at the top. That agreement gives me more confidence that this is a real pattern in the data rather than a quirk of one model. For RuleFit, the effect is very lopsided: for most weeks, a low ratio to the threshold pulls the score down only a little, but in the few weeks where the ratio is high, it pushes the score up strongly (Figure 14). For Negative Binomial, the weather features fill places 3 to 15, led by temperature and humidity two weeks earlier.

4.4 Partial dependence of rainfall

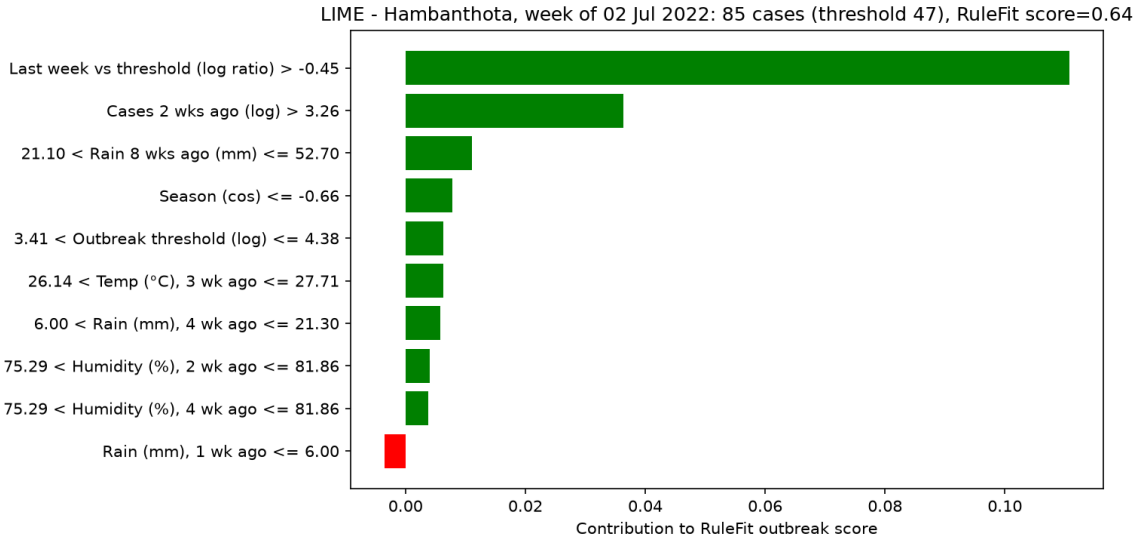
Since rain is what most people associate with dengue, I looked at rainfall in detail. Figure 15 shows how the average prediction changes as each rainfall feature is varied while everything else stays as observed.



Rain from 6–9 weeks earlier is the clearest weather signal in both models. In Negative Binomial, expected cases rise from about 36 to 44 per week across the observed range. XGBoost shows the same rise up to roughly 100 mm and then flattens. That kind of saturation is something a log-linear model cannot represent, which is one reason XGBoost is slightly more accurate. By contrast, rain in the last one or two weeks has almost no effect, and Negative Binomial even gives it a slight negative slope. A plausible explanation is that very heavy recent rain washes away containers and larvae before new adult mosquitoes can emerge.

4.5 LIME: explaining one week

To show how an individual alert can be explained, I chose a real outbreak that RuleFit flagged correctly: Hambanthota in the week of 2 July 2022, with 85 cases against a threshold of 47. LIME fits a simple model around that one week to see which inputs pushed the score up or down. I held the district and monsoon at their real values, so the explanation focuses on the things that change from week to week.



The main reason was that last week's cases were already above about 64% of the threshold. The second was that cases two weeks earlier were also high (more than about 25). Smaller pushes came from moderate rain eight weeks earlier, the July season, temperatures of 26–28 °C three weeks earlier, and humidity of 75–82%. This local explanation matches the global rules in Section 4.2, which is what you would hope to see.

4.6 Does this match what we know about dengue?

Mostly, yes.

Finding from the models	Evidence	What is known about dengue	Does it agree?
Outbreaks build on recent cases	Case-history IRRs of 1.1–2.5; top features in SHAP and RuleFit	The virus passes from person to mosquito to person, so clusters grow over weeks	Yes
Rain acts with a long delay	Rain 6–9 weeks ago is significant (IRR 1.055); correlation peaks at 7–9 weeks	Egg to adult takes 1–2 weeks, then about 8–12 days in the mosquito and 4–7 days in a person	Yes
Humid, warm weeks raise risk 2–4 weeks later	Humidity IRR 1.08, temperature IRR 1.20; RuleFit's humidity rule	Humidity helps mosquitoes survive; warmth shortens the virus's development in the mosquito	Yes
Two peaks each year	Season terms; Figure 2	Peaks follow the south-west and north-east monsoons	Yes
Temperature 1 week ago *lowers* cases (IRR 0.84)	Opposite sign to the 2-week lag	No biological reason to expect this	Partly. Temperatures in neighbouring weeks are strongly correlated, so the model splits one effect into a positive and a negative part. Taken together, the temperature effect is positive.
Recent rain has no clear effect	Rain 1–3 weeks ago, $p > 0.4$	Heavy rain can flush out breeding sites	Plausible

The one caution I would stress is that recent cases dominate everything. The models are mainly learning that trends continue. The weather variables do add early-warning information, but forecasting an outbreak well before the first rise in cases would need field data such as larval surveys.

5. Critical discussion

I think a health office could genuinely use these models as a weekly check on where things are heading, but I want to be upfront about what they can't do. They are better at saying "this is getting worse" than at predicting when an outbreak will start, and they inherit every weakness of passively reported case data.

5.1 Limitations of the models

The most important limitation is the **one-week lag at turning points** (Figure 6). Recent cases are the strongest predictor, so every model, mine and the baselines alike, recognises a surge only after it has started. The early warning is really "risk is rising", not "there will be an outbreak next month".

Negative Binomial is a log-linear model, so it cannot capture effects that level off, such as the rainfall plateau that XGBoost found (Figure 15). That partly explains why it is about one case per week less accurate. Its coefficients also need to be read with care. Weather variables in neighbouring weeks are highly correlated, which is why temperature one week ago and two weeks ago have opposite signs. They are best interpreted as a group rather than one by one.

RuleFit's output is a ranking score, not a true probability. Its best alert cut-off was only 0.046, which would confuse anyone who read it as "4.6% chance". In the app I therefore convert scores into the historical outbreak rate for similar weeks (Section 6). Its recall is also moderate: at the chosen cut-off it catches about half of outbreak weeks, with roughly 1.5 false alarms for every correct alert. Lowering the cut-off would catch more outbreaks at the cost of more false alarms. I believe that trade-off should be set together with health officials, not by me.

5.2 Data quality issues

- **Under-reporting.** Notified cases are mostly people diagnosed in hospital. Mild and symptom-free infections never appear, and how thoroughly cases are reported differs between districts and over time.
- **COVID-19.** My validation years, 2020–21, were unusual. The average was 19 cases per district-week, compared with 32 in the training years, probably because fewer people went to hospital and testing was disrupted. Settings tuned on these years may not be ideal for a normal year.
- **The 2017 epidemic** is an extreme event in the training data. It also pushed the 52-week threshold up for about a year afterwards (Figure 4), so some smaller outbreaks in 2018 were never labelled as outbreaks.
- **Weather from one point per district.** A single centre point cannot represent rainfall across a large or mountainous district such as Ratnapura or Nuwara Eliya. Kalmunai is a health area inside Ampara district, so its weather point partly overlaps with Ampara's.
- **Missing drivers.** Larval survey results, changes in the circulating virus serotype, population density, urbanisation and vector-control activities all affect dengue, but none was available as weekly district data.

5.3 Risks of bias and unfairness

The biggest fairness risk comes from uneven reporting. Districts with stronger surveillance systems, such as Colombo and Gampaha, capture a larger share of their real infections. Rural and Northern districts that report less may therefore look lower-risk than they really are. If the model were used to allocate resources, it could quietly steer them away from the places that already have the least.

The outbreak label has a built-in bias of its own. It is relative to each district's own history, so a district with constantly high dengue is flagged less often than one with a small rise from a low base. "Outbreak" here means "unusual for this district", not "the most cases". This is fair for spotting anomalies, but it should not be used alone to rank districts against each other.

Finally, the district coefficients reflect history, not causes. A high district IRR, such as Nuwara Eliya's 1.82, is not a judgement on the people or local authorities there.

5.4 Real-world impact and ethical considerations

If used carefully, a weekly district forecast that comes with reasons could help Regional Medical Officers start fogging, school clean-ups and hospital preparations a week earlier, and the readable rules make it easier for them to trust or challenge an alert. The ethical risks are just as real:

- **False reassurance.** At the current cut-off, about 46% of outbreak weeks are missed. The tool must support epidemiologists and field surveillance, never replace them.
- **Alert fatigue.** Around 380 false alarms over four years across 26 districts works out to about one every 14 weeks per district. That is manageable, but only if users understand it in advance.
- **Stigma.** Publishing district risk levels openly could stigmatise areas or affect tourism, so outputs should go to health authorities first.
- **Privacy.** I used only aggregated public counts, so no individual can be identified. Moving to finer data, such as individual Medical Officer of Health areas, would need a proper privacy review.
- **Model drift.** Climate change, new serotypes and changes in reporting will all shift the patterns over time. The model should be retrained and re-checked at least once a year.

6. Front-end integration

To make the models usable by someone who is not a data scientist, I built an interactive dashboard in **Streamlit** (`app/streamlit_app.py`). A user picks a district and a week, can change the inputs, and immediately sees the forecast, the outbreak risk, and why the models think so.

6.1 What the user can enter

In the sidebar, the user chooses a **district** and any **week from 2022–2025**, the period the models never saw during training. The scenario inputs are grouped into three sets: recent cases (last week, two weeks ago, four-week average), rainfall (each of the last four weeks, eight weeks ago, and the total from six to nine weeks ago), and temperature and humidity two weeks ago.

When a week is selected, these fields are filled with the values that were actually recorded. The user can then change any of them to explore a **what-if** question, such as "what if it had been 85% humid?", and press *Update prediction*. Features that depend on the inputs, such as the four-week rain total and last week's ratio to the threshold, are recalculated automatically, so the model never receives an inconsistent combination. A small label under the title shows whether the page is showing recorded data or a modified scenario.



Figure 17: the dashboard for Colombo, week of 23 December 2023, showing the headline numbers and the outbreak risk panel.

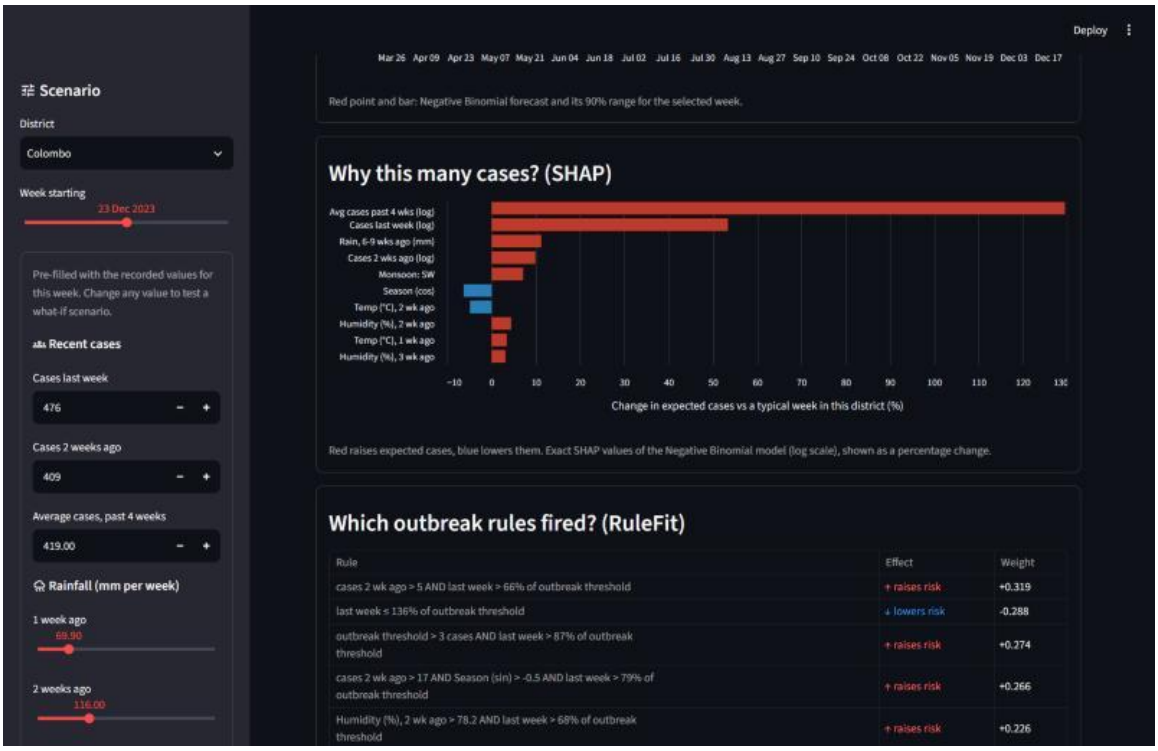


Figure 18: the "Why this many cases?" SHAP panel and the "Which outbreak rules fired?" table for the same week.

6.2 What the user sees

Panel	What it shows	Behind it
Headline numbers	Expected cases, the 90% range, the outbreak threshold, and the cases actually recorded	Negative Binomial mean and quantiles
Outbreak risk	**Low, Moderate or High**, plus a sentence such as "55% of similar weeks in 2007–2021 were outbreaks"	RuleFit score, calibrated against past outbreak rates
History chart	The last 40 weeks of cases and threshold, with this week's forecast and its range	Altair chart
"Why this many cases?"	The ten features that moved the forecast most, as a % change from a typical week in that district	Exact SHAP values from Negative Binomial
"Which outbreak rules fired?"	The RuleFit rules that apply to this scenario, in plain language, marked as raising or lowering risk	RuleFit
Model information	Test-set results, data sources and limitations	–

6.3 Design decisions

A few decisions came from testing the first version of the app and finding it confusing.

My first version showed RuleFit's raw score, which produced messages like "score 0.06 is above the cut-off 0.046". That is meaningless to a health officer. I replaced it with a risk level based on how often weeks with a similar score turned out to be real outbreaks in the past. This is honest about what the score means and easy to read.

The SHAP panel had a similar problem. Compared with the national average week, every Colombo forecast showed recent cases adding "+900%", simply because Colombo is a large district. I changed the comparison to a typical week *in the selected district*, which makes the bars meaningful.

I also added a section called "How the models work overall", with three tabs showing the saved incidence rate ratio, partial dependence and LIME charts. They don't change with the selected week, and the page says so. I kept LIME as one fixed example because running it live would slow the app down.

I kept to standard Streamlit components and gave each chart a short text description. The inputs sit inside a form, so the page only recalculates when the button is pressed, and the models are loaded once and cached so the app stays quick. I tested it on a desktop screen and at phone width (375 px), and it works on both without horizontal scrolling.

To run it locally: `py -m streamlit run app/streamlit_app.py`. A short demo video (3–5 minutes) is included with the submission.

7. Conclusion

I started this project wanting to know whether freely available data could warn of dengue outbreaks in Sri Lanka early, and explain why. After a lot of trial and error with two algorithms from outside our syllabus, my answer is yes, with some caveats. Negative Binomial regression forecast weekly district cases with $R^2 = 0.80$, and its 90% ranges proved reliable (93.5% coverage). RuleFit detected outbreak weeks with a PR-AUC of 0.43, nearly five times the base rate, and had the highest precision of the four classifiers. Both performed close to Random Forest and XGBoost, while being much easier to explain and to question.

What I found most reassuring is that the explanations made biological sense: momentum from recent cases, humid and warm weather two to four weeks earlier, and rain about two months before. The weakness is equally clear: the models mostly confirm a rise that has already started. With larval survey data, serotype information and finer-scale weather, I believe a similar approach could warn before the first cases appear, which is where the real public-health value lies.

On a personal level, the most useful lesson was that careful problem framing mattered more than the choice of algorithm. Defining the outbreak label without leaking the future, splitting the data by time, and adding the threshold-distance feature each changed the results more than switching between models did.

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